LECTURE 3

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Through this talk, we will discuss the power opertations and their relationship with the deformation theory. Our main aim is to prove the cofreeness of the Lubin-Tate theory. We will cover the contents of [BSY22, Section 3]. We also give a more handful proof of this main theorem at the end of the talk.

1. Power operations

Let E be an $\mathbb{E}_\infty$-ring spectrum. We work in the world of E-algebras. The power operations are some natural transformations $\pi_i(-) \to \pi_j(-)$ of functors $\mathrm{hAlg}_E \to \mathrm{Sets}$. In hAlg_E, π_i is corepresented by $E\{t\} = \mathbb{P}(\Sigma^i E) \in \mathrm{hAlg}_E$, where $\mathbb{P}(-) = \bigoplus_{m\geqslant 0} \mathbb{P}_m(-)$ is the free commutative E-Alg monad on hMod_E. By Yoneda, one has the following bijection between sets.

$$\begin{array}{ccc} \{\text{Power operations } \theta : \pi_i(A) \to \pi_j(A)\} & \longleftrightarrow & \pi_j(\mathbb{P}(\Sigma^i E)) \\ \uparrow & & \uparrow \\ \{\text{Additive power operations } \theta : \pi_i(A) \to \pi_j(A)\} & \longleftrightarrow & \{\text{Primitive elements in } \pi_j(\mathbb{P}(\Sigma^i E))\} \end{array}$$

The examples of power operations include the Steenrod operation and Adams operations on p-complete K-theory. Through this talk we focus on $E = E(k)$ for a fixed prime p and a height n. In [Rez09], Rezk constructed an algebraic approximation of $\mathbb{P}$, which is a monad on Mod_{E_*} denoted by $\mathbb{T}$.

The construction of Rezk can be concluded as follows. We consider the equivalence induced by π_*

$$\mathrm{hMod}_E^{ff} \xrightarrow{\simeq} \mathrm{Mod}_{E_*}^{ff}$$

If we restrict to $K(n)$-local finite free E-algebras, then $\mathbb{P}_m$ becomes a submonad. One can define the desired $\mathbb{T}_m$ as the left Kan extension

$$\begin{array}{ccccc} \mathrm{hMod}_E^{ff} & \hookrightarrow & \mathrm{hMod}_E & \xrightarrow{\pi_* L_{K(n)} \mathbb{P}_m} & \mathrm{Mod}_{E_*} \\ \downarrow \simeq & & \downarrow \pi_* & \nearrow_{\mathbb{T}_m} & \\ \mathrm{Mod}_{E_*}^{ff} & \hookrightarrow & \mathrm{Mod}_{E_*} & & \end{array}$$

2. Main theorem

We consider the forgetful functor $U_{\mathbb{T}}$ from $\mathbb{T}$-algebras to E_0-algebras, which admits both a left adjoint $F_{\mathbb{T}}$ and a right adjoint $W_{\mathbb{T}}$.

$$\mathrm{Alg}_{\mathbb{T}} \underset{W_{\mathbb{T}}}{\overset{F_{\mathbb{T}}}{\underset{\longrightarrow}{\overset{\longleftarrow}{U_{\mathbb{T}}}}}} \mathrm{CRing}_{E_0}$$

Applying the above adjunction to the height-1 case and considering the classical adjunction

$$(-/p)^\sharp : \mathrm{CRing} \rightleftharpoons \mathrm{Perf}_{\mathbb{F}_p} : \mathrm{incl}$$

We obtain an adjunction

$$(U(-)/p)^\sharp : \mathrm{CRing}_\delta \rightleftharpoons \mathrm{Perf}_{\mathbb{F}_p} : \pi_0 E(-)$$

Our main theorem generalizes this observation to arbituary height.

Theorem 2.1. *We fix a height n and let $\mathfrak{m} = \mathfrak{m}_n$ be the Landweber ideal. Then there is an adjunction*

$$(U(-)/\mathfrak{m})^\sharp : \mathrm{Alg}_{\mathbb{T}} \rightleftharpoons \mathrm{Perf}_{\mathbb{F}_p} : \pi_0 E(-)$$

where the right adjoint $\pi_0 E(-)$ is fully faithful.

Definition 2.2. Let $\mathbb{1}$ be the symmetric unit in a stable symmetric monoidal category $\mathcal{C}$. To each $P \in \pi_0(\mathbb{1}_{h\Sigma_r})$, we can associate a power operation of weight r, which takes $x \in \pi_0(R)$ to

$$\mathbb{1} \xrightarrow{P} \mathbb{1}_{h\Sigma_r} \to \oplus \mathbb{1}_{h\Sigma_r} \simeq \mathbb{1}\{t\} \to R$$

This induces a map $P_r : \pi_0 R \to [B\Sigma_r, R]$. We call it the total power operation of weight r.

Let A be a perfect k-algebra. The underlying set of $W_{\mathbb{T}}(A)$ can be identified with the set of A-valued functions on E_0-mod of power operations. Let $ev_A : A \to W_{\mathbb{T}} U_{\mathbb{T}}$. Consider the following diagram

$$\begin{array}{ccc} & W_{\mathbb{T}} U_{\mathbb{T}} \pi_0 E(A) & \\ {\scriptstyle ev_A}\nearrow & & \searrow{\scriptstyle W_{\mathbb{T}}(-/\mathfrak{m})} \\ \pi_0 E(A) & \xrightarrow{\overline{ev}_A} & W_{\mathbb{T}}(A) \end{array}$$

we will prove that $\overline{ev}_A$ is an equivalence for every $A \in \mathrm{Perf}_k$, then everything follows immediately.

3. The proof

3.1. **The Witt filtration.** For $\lambda \in \mathbb{T}(E_0)$, we have an evaluation map $\lambda^* : W_{\mathbb{T}}(A) \to A$. We define an ideal

$$W_{\mathbb{T}}^{\geqslant r}(A) = \{v \in W_{\mathbb{T}}(A) | \lambda^* v = 0 \text{ for any power operation } \lambda \text{ with weight less than } p^r\}$$

Definition 3.1. We let $W_{\mathbb{T}}^{\leqslant r}(A) = W_{\mathbb{T}}(A)/W_{\mathbb{T}}^{\geqslant r+1}(A)$. This defines a filtration which we call the **Witt filtration**.

$$W_{\mathbb{T}}(A) \to \cdots \to W_{\mathbb{T}}^{\leqslant r}(A) \to \cdots \to W_{\mathbb{T}}^{\leqslant 1}(A) \to W_{\mathbb{T}}^{\leqslant 0}(A) = A$$

Strickland shows that we have an equivalence of underlying sets

$$W_{\mathbb{T}}(A) \cong \mathrm{Hom}_{E_0}(\mathbb{T}(E_0), A) \cong \prod_{r \geqslant 0} \overline{d}(r)$$

where $\overline{d}(r)$ denotes the Gaussian polynomial coefficience.

Proposition 3.2 ([BSY22]; Proposition 3.27)**.** *For any $A \in \mathrm{Alg}_{E_0}$, we have an equivalence of graded pieces*

$$W_{\mathbb{T}}^{=r}(A) = W_{\mathbb{T}}^{\geqslant r}(A)/W_{\mathbb{T}}^{\geqslant r+1}(A)$$

which fits into the commutative diagram

$$\begin{array}{ccc} W^{\geqslant r}(A) & \longrightarrow & W^{=r}(A) \\ \cong\downarrow & & \downarrow \\ \prod_{i \geqslant r} \overline{d}(i) & \longrightarrow & \overline{d}(r) \end{array}$$

The A-mod structure is not obvious. Instead, one can show that the A-operation is given by $[a]v = a^{p^r}v$. As a result, $W_{\mathbb{T}}^{=r}(A)$ becomes a free A-module of rank $\overline{d}(r)$.

One can use this to form the base-change for reduction. We have the following consequence which allows us to reduce to k.

Theorem 3.3. *If $\overline{ev}_k : \pi_0 E(k) \to W_{\mathbb{T}}(k)$ is an isomorphism, then $\overline{ev}_A$ is an isomrophism for every $A \in \mathrm{Perf}_k$.*

Proof. We have the following commutative diagram, where the vertical arrows are completion maps.

$$\begin{array}{ccc} W(A) \otimes_{W(k)} \pi_0 E(k) & \longrightarrow & W(A) \otimes_{W(k)} W_{\mathbb{T}}(k) \\ \wedge\downarrow & & \downarrow \\ \pi_0 E(A) & \longrightarrow & W_{\mathbb{T}}(A) \end{array}$$

It follows from the observation that the $\mathfrak{m}$-adic topology and W-topology coincide. □

3.2. **Surjectivity.** We want to show the following.

Proposition 3.4. *For every weight r, the mod $\mathfrak{m}$ total power operation $\overline{\tau}_r : \pi_0 E(k) \to E^0(B\Sigma_{p^r})/(I_{tr}, \mathfrak{m})$ is surjective.*

In order to prove the surjectivity, we need to use the Moduli interpretion of power operations.

We consider the moduli problem. We consider the functor LT from complete local rings to groupoids, sends R to deformations of G_0 over k. This is correpresented by a Lubin Tate ring $\mathcal{O}_{\mathrm{LT}}$, which non-cannonically isomorphic to $W(k)[\![u_1, \cdots, u_{n-1}]\!]$.

Theorem 3.5 ([AHS04]; Proposition 12.12)**.** *Let E be the Lubin-Tate theory attached to a height n formal group $\mathbb{G}_0$ over a perfect field k. Let $\varphi^r : \mathbb{G}_0 \to \mathbb{G}_0^{(r)}$ be the r-fold Frobenius isogeny. Then there is an isomorphism of schemes*

$$\mathrm{Def}(\varphi^r) \cong \mathrm{Spf}(E^0(B\Sigma_{p^r})/I_{tr})$$

Therefore, in order to prove Proposition 3.4, it suffices to show we have a close immersion

$$\mathrm{Def}(\varphi^r) \xrightarrow{(\sigma.\tau)} \mathrm{Def}(\mathbb{G}_0) \times_{\mathrm{Spf}W(k)} \mathrm{Def}(\mathbb{G}_0^{(r)})$$

which follows from a more general fact in algebraic geometry.

Proposition 3.6. *Let $q_0 : \mathbb{G}_0 \to \mathbb{G}_0'$ an isogeny of formal groups over l. Then*

$$\mathrm{Def}(q_0) \xrightarrow{(\sigma.\tau)} \mathrm{Def}(\mathbb{G}_0) \times_{\mathrm{Spf}W(k)} \mathrm{Def}(\mathbb{G}_0')$$

is a closed immersion.

3.3. **Complete the proof.** We can complete the proof that $\overline{ev}_A : A \to W_{\mathbb{T}}U_{\mathbb{T}}A$ is an isomorphism.

Definition 3.7. A p-derivation is $\theta \in \mathbb{T}(E_o)_p$ such that for any $\mathbb{T}$-algebra B and $x, y \in B$, $\theta(x+y) = \theta(x) + \theta(y) + (x^p + y^p - (x+y)^p)/p$.

Using induction for height and regard the whole paper as inductive, one can show the following

Lemma 3.8 ([BSY22]; Proposition 3.45)**.** *For any nonezero $y \in E_0$, there is an additive power operation Q together with a p-derivation θ such that $\theta^i Q(y)$ is nonzero.*

Therefore, one can conclude the results by assemble the consequences for $\overline{\tau}_r : \pi_0 E(k) \to E^0(B\Sigma_{p^r})/(I_{tr}, \mathfrak{m})$.

Theorem 3.9. *The evaluation map $\overline{ev}_k : E_0(k) \to W_{\mathbb{T}}(k)$ is an isomorphism.*

Proof. The evaluation map can be interpreted as evaluation functions in power operations. The injective part comes from Lemma 3.8. The surjective part is from Proposition 3.4 and additional length arguments. □

3.4. **Akhil's proof.** The proof of [BSY22] is somehow technical. Akhil gives a more concise proof.

Let Ψ be the power operation that corresponds to the isogeny $[p]$ through Strickland's correspondence. This implies that Ψ is central in additive power operations. We say a $\mathbb{T}$-algebra R is perfect if $\Psi : R \to R$ is an isomorphism. We can obtain a colimit perfection of $R \in \mathrm{Alg}_{\mathbb{T}}$.

$$R^{\mathrm{Perf}} = \mathrm{colim}(R \xrightarrow{\Psi} R \xrightarrow{\Psi} R \xrightarrow{\Psi} \cdots)$$

We have an adjunction

$$(\hat{-})^{\mathrm{Perf}} : \mathrm{Alg}_{\mathbb{T}} \rightleftharpoons \mathrm{PerfAlg}_{\mathbb{T}}^{\wedge} : \mathrm{forget}$$

It suffices to prove that we have an equivalence

$$(-)/\mathfrak{m} : \mathrm{PerfAlg}_{\mathbb{T}}^{\wedge} \xrightarrow{\simeq} \mathrm{Perf}_k$$

The essentially full is clear. To see that the functor is fully faithful, the key observation is the following

Lemma 3.10. *The functor $(\hat{-})^{\mathrm{Perf}} : \mathrm{Alg}_{\mathbb{T}} \to \mathrm{PerfAlg}_{\mathbb{T}}^{\wedge}$ sends R to $E_0(A)$ for some perfect k-algebra A.*

Proof idea. We can define $A := (\hat{R}//\mathfrak{m})^{\mathrm{Perf}}$ and check that $\hat{R}^{\mathrm{Perf}} \simeq E_0(A)$ as $\mathbb{T}$-algebras. □

Therefore, every perfect $\mathfrak{m}$-complete $\mathbb{T}$-algebra is of the form $E_0(A)$. By formal étaleness we have

$$\mathrm{Alg}_{\mathbb{T}}(E_0(A_1), E_0(A_2)) \subset \mathrm{Alg}_{E_0}(E_0(A_1), E_0(A_2)) = \mathrm{Alg}_k(A_1, A_2)$$

which admits a section sending $A_1 \to A_2$ to $E_0(A_1) \to E_0(A_2)$.

References

[AHS04] Matthew Ando, Michael J Hopkins, and Neil P Strickland. "The sigma orientation is an H_∞ map". In: *American journal of mathematics* 126.2 (2004), pp. 247–334 (cit. on p. 3).

[BSY22] Robert Burklund, Tomer M Schlank, and Allen Yuan. "The chromatic nullstellensatz". In: *arXiv preprint arXiv:2207.09929* (2022) (cit. on pp. 1, 3, 4).

[Rez09] Charles Rezk. "The congruence criterion for power operations in Morava E-theory". In: *arxiv:0902.2499v2* (2009) (cit. on p. 1).